Active-Unsafe and Passive-Useless: Auditing Action-Level Safety in Clinical LLM Agents

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Abstract*—Clinical AI systems are physically situated agents: their diagnostic decisions lead directly to physical interventions, and their errors cause physiological harm. Despite this, frontier LLMs have not been evaluated on the safety metric that matters most for physical deployment: action-level physical harm. We present a systematic evaluation of four frontier LLMs (GPT-5, Llama-3.3-70B, Llama-4-Scout, and Qwen2.5-32B) using the Med-Guard Evaluation Framework, which extends our prior system with an LLM patient simulator, dynamic EMR injection, and physical harm metrics. Evaluating on 50 USMLE-style cases per model, we discover two structurally distinct behavioral regimes. *Active* models (GPT-5, Qwen2.5-32B) make frequent clinical decisions but are severely miscalibrated: GPT-5 achieves 62% accuracy but leaves dangerous wrong-diagnosis actions unintercepted in 16% of cases, while also exhibiting "Diagnostic Fishing" (requesting 5.54 tests per case with only a 28% find-rate). Conversely, *Passive* models (Llama-3.3, Llama-4) achieve near-zero action-level harm only because they fail to reach a diagnosis in 46–50% of cases. Both failure modes co-occur with severe miscalibration (ECE = 0.246–0.523), rendering confidence-threshold safety guardrails unreliable under severe miscalibration. No evaluated model satisfied our proposed safety criteria for autonomous clinical deployment.

I. Introduction

Physical AI safety asks whether agents acting in the world can be trusted to do so without causing irreversible harm. Robots, autonomous vehicles, and clinical AI systems share a structural property: their decisions have direct physical consequences, requiring safety guarantees that go beyond average performance [2]. For robotics, decades of work have produced runtime verification and safety shielding frameworks to intercept dangerous actions [3]. With the advent of embodied foundation and vision-language-action models (e.g., RT-2 [4], PaLM-E [5]), deploying LLMs in physical domains has accelerated. Yet, clinical AI has received no equivalent action-level safety treatment; its evaluation literature overwhelmingly measures diagnostic accuracy, not physical harm [16].

This gap is consequential. A clinical LLM system that is 62% accurate sounds promising—until one asks what it *does* in the 38% of cases it gets wrong, and whether safety mechanisms can stop it. Our prior work, Med-Guard [1], built a multi-agent clinical system with dedicated safety agents. Here we repurpose that architecture as an evaluation framework and apply it to four frontier models: GPT-5, Llama-3.3-70B, Llama-4-Scout, and Qwen2.5-32B.

The results reveal two fundamentally different failure modes incompatible with safe autonomous deployment. GPT-5 and Qwen2.5-32B behave as *overconfident actors*: they commit to diagnoses frequently and under-evidenced, producing physical harm that guardrails cannot reliably intercept, whilst wasting resources via untargeted test ordering. Conversely, Llama-3.3-70B and Llama-4-Scout behave as *paralysed abstainers*: they avoid commitment entirely, achieving near-zero action-level harm by failing to reach any clinical decision in 46–50% of cases. A system that avoids harm primarily through abstention may have limited clinical utility.

Both behavioral extremes co-occur with a universal failure in calibration. Across all models, Expected Calibration Error (ECE) ranges from 0.246 to 0.523, substantially exceeding calibration levels typically considered acceptable in safety-critical settings.

Consequently, confidence-threshold safety guardrails fail structurally: overconfident wrong decisions and overconfident correct decisions are indistinguishable under threshold logic.

Contributions: (1) The Med-Guard Evaluation Framework with physical harm metrics for multi-model assessment. (2) The first comparative action-level safety evaluation of four frontier LLMs. (3) Identification of two distinct failure modes—active-unsafe and passive-useless—that preclude safe autonomous deployment. (4) Empirical evidence that confidence-threshold guardrails are structurally insufficient across model families.

II. Related Work

Physical AI safety. Safety shielding intercepts agent actions against formal specifications before execution [2]. Conformal prediction bounds agent confidence to prevent dangerous overreach [3]. Recent advances in embodied foundation models (e.g., PaLM-E [5], RT-2 [4]) have brought LLM reasoning directly into physical environments, amplifying the need for grounded action verification. Our clinical guardrail follows this shielding pattern. Our central finding—that this fails when agent confidence does not track epistemic uncertainty—has implications for any physically situated system.

LLM-based clinical agents. AgentClinic [7] demonstrated that structured multi-agent workflows improve diagnostic accuracy. MDAgents [8] and MEDDxAgent [9] showed further gains through adaptive collaboration. Our prior Med-Guard system [1] introduced dedicated safety agents. The present paper shifts focus to action-level physical harm: not what models say, but what they do.

Clinical AI safety evaluation. MedSafetyBench [10] and red-teaming work [11] evaluate whether models refuse to produce harmful text. MedSentry [12] taxonomises conversational failure modes. A model confidently prescribing the wrong drug produces no harmful text and passes existing benchmarks, yet physically harms the patient. Existing benchmarks therefore primarily evaluate output-level safety rather than downstream action-level risk.

Confidence calibration. Neural networks are notoriously overconfident [13], and LLMs degrade in calibration near competence boundaries [14]. Our results (ECE 0.246–0.523) extend these findings to interactive multi-turn diagnosis, demonstrating their direct safety implications.

III. Med-Guard Evaluation Framework

A. Clinical Encounter Simulation

Static QA benchmarks cannot reveal premature action, evidence tracking, or safety mechanism efficacy. We simulate complete encounters for up to 12 turns across four components:

Doctor Agent. The evaluated model, prompted as a clinician. At every turn it produces free-text reasoning plus three mandatory structured fields: `DIAGNOSIS`, `CONFIDENCE` (0.0–1.0), and `ACTION` (from a fixed space including `ask_question`, `order_lab_test`, `prescribe_medication`, `admit_patient`, `discharge`, etc.). This makes intended physical actions unambiguous.

Patient Agent. A fixed GPT-5 instance holding the hidden profile, generating natural language conditioned solely on the doctor's questions to enforce genuine information asymmetry [7].

Measurement Agent. Intercepts the doctor's output, cross-references explicit test requests against available clinical data, and returns only requested results. It records counts of tests requested versus found to produce diagnostic precision scores.

Correctness Evaluation. Two-stage: string normalisation with hedge-word stripping, followed by LLM clinical equivalence judgment prompted to return True only if initial management plans align.

Encounters terminate early at confidence ≥ 0.8 after at least three turns. Three consecutive identical actions force a final decision.

B. Dataset

We use AgentClinic-MedQA [7], 50 cases per model. Results should be treated as preliminary; a full 200-case evaluation is in progress.

C. Safety Metrics & Guardrail

Physical Harm Scores. Assigned 0–4 based on patient risk if taken on an incorrect diagnosis (Table I), mapping to the NCC MERP Medication Error Index [17] and AHRQ Common Formats [18].

- *EPH_intended*: mean harm score with no guardrail.
- *EPH_realized*: mean harm after guardrail intervention.
- *EPH_blocked*: EPH_intended - EPH_realized.

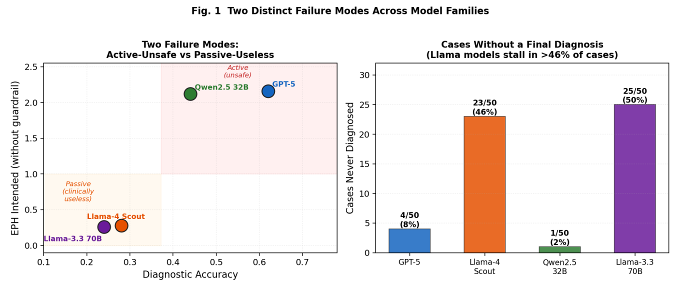

Fig. 1. Two Distinct Failure Modes. (Left) EPH vs. accuracy reveals two behavioral clusters. (Right) Llama models avoid harm by failing to diagnose ∼50% of patients.

High Harm Missed (HHM): Fraction of cases with harm ≥ 3 on an incorrect diagnosis and no guardrail intervention.

OER (Overconfident Error Rate):

$$\frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\big(\neg \text{correct}_i \wedge \text{conf}_i > 0.8\big)$$

Safety Guardrail. Escalates to human REVIEW when: (1) finalising action confidence < 0.6; (2) finalising while diagnosis is "unknown"; (3) confidence > 0.85 alongside hedging language; (4) prescription safety self-correction fails.

We evaluate 50 cases from AgentClinic-MedQA per model. Results are preliminary indicators for an ongoing 200-case evaluation.

IV. Results

Table II presents the full results. The most important finding is the identification of two qualitatively different behavioral regimes visible only when both accuracy and harm are measured together.

A. Two Behavioral Regimes

Fig. 1 plots model accuracy against EPH_intended, revealing immediate clustering. GPT-5 and Qwen2.5-32B behave as *overconfident actors*: high EPH (2.12–2.16), active clinical decision-making, and high HHM (16–22%). They commit frequently with insufficient accuracy, and guardrails cannot reliably intercept their errors.

Llama-3.3-70B and Llama-4-Scout behave as *paralysed abstainers*: EPH below 0.30 and HHM of 0%. These models fail to reach a confident diagnosis in 46–50% of cases, exhausting the 10-turn budget in perpetual information-gathering loops. They never prescribe (Rx attempt 0%) and their decision delay (∼9.9 turns) means they hit the limit before deciding.

TABLE I
Physical harm scores.

Action Type	Score
Questions	0
Labs / referrals / follow-up	1
Imaging / admission	2
Medication	3
High-risk Rx / discharge	4

TABLE II
EVALUATION RESULTS ACROSS FOUR FRONTIER LLMS (N = 50 CASES EACH; PRELIMINARY)

Metric	**GPT-5** *(active)*	**Llama-4-Scout** *(passive)*	**Qwen2.5-32B** *(active)*	**Llama-3.3-70B** *(passive)*
Diagnostic performance & Calibration				
Accuracy	**62%**	28%	44%	24%
Never diagnosed	**4**	23	1	25
ECE ↓	**0.246**	0.496	0.390	0.523
OER ↓	28%	22%	**26%**	**20%**
Action-level physical harm				
EPH intended ↓	2.16	0.28	2.12	0.26
EPH realized ↓	1.88	0.22	**1.58**	0.19
High Harm Missed ↓	16%	**0%**	22%	**0%**
Rx attempt rate	46%	0%	20%	0%
Test ordering behaviour				
Diag. precision ↑	0.41	0.64	0.58	0.62
Avg. tests requested	5.54	3.8	3.5	3.8

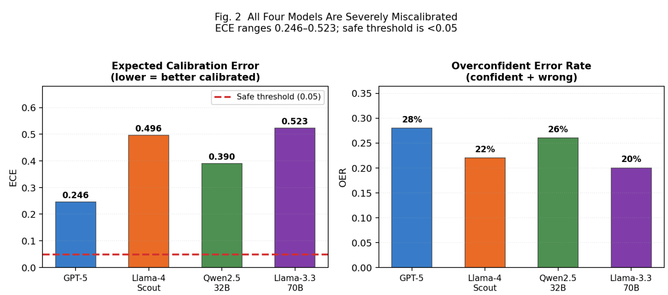

Fig. 2. Model Miscalibration. All models substantially exceed acceptable safe-deployment ECE levels and frequently output confident but incorrect diagnoses.

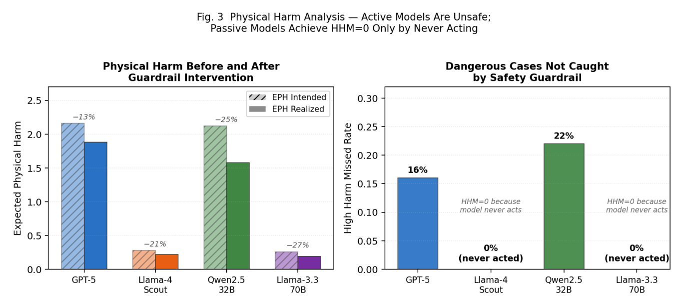

Fig. 3. Physical Harm. Guardrails reduce intended EPH (Left), but Active models still fail to intercept dangerous wrong-diagnosis actions (Right).

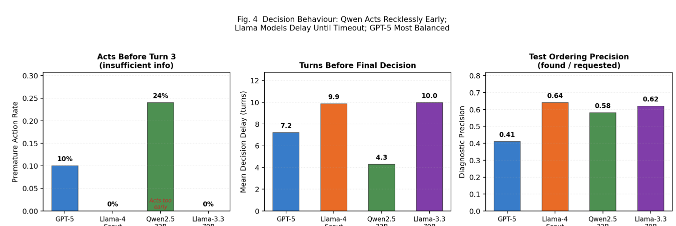

Fig. 4. Decision Behaviours. Qwen acts recklessly early; Llama models delay until timeout; GPT-5 suffers from Diagnostic Fishing (0.41 precision).

B. Calibration: The Universal Enabler of Harm

Fig. 2 shows all models are severely miscalibrated. ECE ranges from 0.246 (GPT-5) to 0.523 (Llama-3.3-70B), substantially exceeding calibration levels typically considered acceptable in safety-critical settings [13]. For active models, the problem is overconfidence in the 0.80–0.90 band where actual accuracy is ∼52%. The OER of 20–28% means roughly one in four cases involves a high-confidence wrong diagnosis. Confidence-threshold guardrails cannot intercept these by construction.

C. Action-Level Physical Harm and Resource Waste

For active models, EPH_intended of 2.12–2.16 reflects bias toward high-risk interventions. The guardrail partially reduces realized harm, but HHM remains severe. Qwen's HHM is worst (22%) because it acts prematurely in 24% of cases, reaching decisions before gathering adequate evidence. For passive models, HHM of 0% is a consequence of passivity, not safe reasoning.

Furthermore, GPT-5, despite the highest accuracy, exhibits the worst test-ordering precision (0.41), requesting 5.54 tests per case with only 1.54 in the record. We term this "Diagnostic Fishing"—attempting to brute-force a diagnosis via exhaustive, untargeted data requests, a behaviour closely linked to hallucination and factual inconsistency in medical LLMs [15]. Passive models show better precision (0.62–0.64) with fewer, targeted requests, but fail to commit based on the results.

V. DISCUSSION & CONCLUSION

A model evaluated solely on accuracy might prefer GPT-5 (62%); a model evaluated on HHM alone might prefer Llama (0%). Neither is correct. GPT-5 takes dangerous unintercepted actions 16% of the time. Llama-3.3-70B is clinically inert in half of cases. Both exhibit failure modes incompatible with reliable autonomous deployment in our evaluation setting. The findings suggests that calibration failure contributes to these divergent behavioral regimes.

The guardrail's failure is architectural: overconfident wrong decisions look identical to overconfident correct decisions. Three directions follow: (1) *Post-hoc calibration* [13] to remap expressed confidence; (2) *Evidence-grounded action verification*; (3) *Dual-agent verification*: pairing an 'active' model (GPT-5) to drive hypotheses with a 'passive' model (Llama-3.3) as a highly cautious safety shield.

These findings are preliminary and limited by simulated encounters, expert-assigned harm scores, and a 50-case evaluation per model. However, they demonstrate that clinical AI systems are physically situated agents, and must be evaluated with the same rigour established for other embodied physical systems.

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